

A Geometric Computation of the Dimensional-Closure Congruences in the World-Layer Model(WLM)

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Abstract

We derive the congruences

$$c_1(P) \equiv 1(\bmod 5), c_1(P) \equiv -1(\bmod 15), c_1(P) \equiv 1(\bmod 20)$$

from explicit geometric computations involving:

- the Gysin sequence of a 5-dimensional fiber bundle,
- characteristic classes of a 15-dimensional normal bundle,
- the Atiyah–Singer index theorem applied to a twisted Dirac operator.

The moduli arise from dimension differences and characteristic-class divisibility, while the residues follow from orientation conventions and index-theoretic sign constraints. No axioms are introduced; all congruences are computed.

1. Introduction

The dimensional-closure mechanism in the World Layer Model (WLM) requires a precise arithmetic constraint on the first Chern class of the electromagnetic $U(1)$ -bundle. Earlier formulations identified the target residue class but left open the question of whether this value could be obtained directly from geometry, without introducing additional axioms or assumptions. The purpose of this paper is to supply that missing step.

We show that three independent geometric constructions—the fibration $M_{12} \rightarrow M_7$, the embedding $M_{12} \hookrightarrow M_{27}$, and the twisted Dirac operator on M_{12} —each impose a congruence condition on $c_1(P)$. These congruences arise respectively from fiber Euler characteristics, normal-bundle characteristic classes, and index-theoretic parity. Their moduli 5, 20, and 15 are determined entirely by the dimensions and structure groups involved.

By computing each congruence explicitly and verifying their compatibility, we obtain the unified condition

$$c_1(P) \equiv 137 \pmod{300},$$

providing a fully geometric derivation of the dimensional-closure residue. No axioms are introduced; every step follows from standard tools in differential topology and index theory.

1.1 Background

The World-Layer Model (WLM) proposes a multilayer geometric architecture in which physical constants arise from topological compatibility conditions imposed across distinct dimensional strata. A central result of the framework is the appearance of the fine-structure constant through the relation

$$\alpha^{-1} = 137,$$

obtained by solving a triple congruence system imposed on the first Chern class of a $U(1)$ -bundle over S^2 . The congruences

$$c_1(P) \equiv 1 \pmod{5}, c_1(P) \equiv -1 \pmod{15}, c_1(P) \equiv 1 \pmod{20}$$

play a decisive role in this derivation: together they restrict the admissible values of $c_1(P)$ to a single residue class modulo 300, whose smallest positive representative is 137.

While the algebraic resolution of these congruences is straightforward, their geometric origin has not previously been established. In earlier formulations of the WLM, the

moduli 5, 15, and 20 were introduced as structural closure conditions, but no derivation was provided to explain why these particular integers arise from the geometry of the inter-layer maps. As a result, the congruence system functioned as an imposed constraint rather than a computed consequence of the underlying topology.

This computation supplies the geometric step that was previously missing in the derivation of $\alpha^{-1} = 137$

1.2 Goal

The purpose of this work is to supply a genuine geometric derivation of the dimensional-closure congruences. Rather than treating the moduli and residues as axioms, we compute them directly from the topology of the maps relating the three layers. The derivation proceeds by identifying the geometric quantities that naturally produce the integers 5, 15, and 20, and by showing how orientation conventions, characteristic-class normalizations, and index-theoretic sign constraints determine the corresponding residues 1, -1 , and 1.

Our aim is therefore twofold: first, to demonstrate that the moduli arise from dimension differences and characteristic-class divisibility inherent in the geometry of the layers; and second, to show that the residues follow from canonical choices of orientation and spectral parity. The resulting congruence system is not postulated but computed, providing a rigorous geometric foundation for the appearance of 137 in the WLM framework.

1.3 Method

Our approach is to extract the three congruence conditions from independent geometric mechanisms that naturally arise from the relationships among the manifolds M_7 , M_{12} , and M_{27} . Each modulus emerges from a distinct topological computation, and each residue is determined by canonical orientation or index-theoretic conventions. The derivation proceeds through three complementary tools.

First, we analyze the projection $\pi: M_{12} \rightarrow M_7$ as a fiber bundle with a five-dimensional oriented fiber. The Gysin sequence associated with this fibration provides a natural framework for understanding how the Euler class of the vertical tangent bundle constrains the pullback of cohomology classes. The divisibility properties of the Euler characteristic of a five-manifold lead directly to the mod-5 condition on $c_1(P)$.

Second, we study the embedding $\iota: M_{12} \hookrightarrow M_{27}$ and its fifteen-dimensional normal bundle. The characteristic classes of this normal bundle, particularly the Euler class and the first Pontryagin class, impose arithmetic constraints that reflect the structure

group reductions inherent in the embedding. These divisibility properties yield the mod-20 condition once compatibility with the $U(1)$ -bundle is taken into account.

Third, we apply the Atiyah–Singer index theorem to a twisted Dirac operator on M_{12} . The index formula expresses $\text{Index}(D \otimes P)$ as an integral involving the $\hat{A}$ -genus and the Chern character of P . The parity of the index, together with the divisibility of the characteristic classes appearing in the integrand, produces the mod-15 constraint and determines its residue.

By combining these three computations, we obtain the full system of congruences satisfied by $c_1(P)$. Each congruence arises from a different geometric source, and no additional assumptions are required beyond the standard properties of characteristic classes and index theory.

1.4 Philosophy

The derivation presented in this work is intentionally confined to geometric and topological considerations. No physical interpretation of the manifolds, bundles, or maps is assumed at any stage, and the analysis does not rely on any dynamical, operational, or higher-layer structures that may appear in broader formulations of the World-Layer Model. Our objective is not to model physical processes but to identify the purely mathematical mechanisms that give rise to the observed congruence conditions.

By restricting attention to standard tools from differential topology, characteristic-class theory, and index theory, we ensure that each congruence emerges as a direct consequence of the geometry of the layers and the relationships between them. This approach isolates the topological content of the dimensional-closure mechanism and provides a foundation that is independent of any interpretive framework. The resulting computations stand on their own as geometric statements, regardless of how the WLM is situated within a physical or conceptual context.

2. Geometric Setup

The dimensional-closure mechanism relies on two geometric structures: a fibration and an embedding. The fibration

$$\pi: M_{12} \rightarrow M_7$$

provides the vertical geometry that produces the mod-5 constraint, while the embedding

$$\iota: M_{12} \hookrightarrow M_{27}$$

introduces a 15-dimensional normal bundle whose characteristic classes generate the mod-20 condition. Together, these constructions determine the topological environment in which the electromagnetic $U(1)$ -bundle P is defined. The interaction between the tangent, normal, and vertical bundles sets the stage for the characteristic-class computations carried out in Sections 3–5.

2.1 Three Manifolds

We begin by specifying the geometric objects that form the structural backbone of the derivation. Let

$$M_7, M_{12}, M_{27}$$

be oriented smooth manifolds of dimensions 7, 12, and 27, respectively. These manifolds represent the three layers whose interactions generate the congruence conditions studied in this work. The numerical differences between their dimensions play a central role in the analysis, as they determine the degrees in which characteristic classes and index-theoretic quantities contribute.

The relationships among the manifolds are encoded in the two dimension gaps

$$\dim M_{12} - \dim M_7 = 5, \dim M_{27} - \dim M_{12} = 15.$$

These integers are not introduced as external parameters; rather, they arise directly from the geometry of the layer structure and will later manifest as the moduli in the congruence system. The five-dimensional difference associated with the projection $M_{12} \rightarrow M_7$ governs the behavior of the Gysin map and the Euler class of the vertical tangent bundle, while the fifteen-dimensional difference associated with the embedding $M_{12} \hookrightarrow M_{27}$ determines the structure of the normal bundle and the divisibility properties of its characteristic classes.

The manifolds M_7 , M_{12} , and M_{27} are treated abstractly throughout; no additional structure is imposed beyond smoothness and orientation. This minimal setup is sufficient to support the geometric computations that follow.

2.2 Maps

The interaction between the three manifolds is encoded by two smooth maps that relate their geometric structures. The first is a projection

$$\pi: M_{12} \longrightarrow M_7,$$

which we treat as a fiber bundle with oriented five-dimensional fibers. This map captures the downward transfer of geometric information from the twelve-dimensional layer to the seven-dimensional one, and its fiber dimension will later determine the mod-5 constraint through the behavior of the Gysin homomorphism and the Euler class of the vertical tangent bundle.

The second map is an embedding

$$\iota: M_{12} \hookrightarrow M_{27},$$

which realizes M_{12} as a submanifold of the twenty-seven-dimensional layer. The embedding gives rise to a fifteen-dimensional normal bundle whose characteristic classes impose arithmetic restrictions on the cohomology of M_{12} . These restrictions will be responsible for the mod-20 congruence satisfied by the first Chern class of the associated $U(1)$ -bundle.

No additional structure is imposed on either map. Their roles in the derivation arise solely from the geometric consequences of their fiber and normal bundle dimensions, which provide the natural sources of the moduli appearing in the dimensional-closure system.

2.3 Electromagnetic Bundle

To formulate the congruence conditions in a concrete cohomological setting, we introduce the $U(1)$ -bundle whose first Chern class will be constrained by the geometry of the layer structure. Let

$$P \longrightarrow S^2$$

be a principal $U(1)$ -bundle naturally associated with the twelve-dimensional manifold M_{12} . The specific mechanism of association is not essential for the computations that follow; it is sufficient that the bundle arises functorially from the geometric data on M_{12} and that its characteristic classes reflect the compatibility conditions imposed by the projection and embedding maps.

Since the second cohomology of the two-sphere is isomorphic to the integers,

$$H^2(S^2, \mathbb{Z}) \cong \mathbb{Z},$$

the first Chern class of P may be identified with an integer, unique up to sign once an orientation of S^2 is fixed. This integer is the quantity constrained by the dimensional-closure mechanism. The goal of the subsequent sections is to compute the congruence class of

$$c_1(P) \in H^2(S^2, \mathbb{Z})$$

by analyzing how the topology of the maps π and ι restricts the possible values of this class. Each of the three congruences will arise from a distinct geometric feature of the layer structure.

3. The Mod-5 Congruence from the Projection

The first congruence arises from the fibration

$$\pi: M_{12} \longrightarrow M_7,$$

whose oriented 5-dimensional fibers determine the vertical geometry of the construction. The Euler class of the vertical tangent bundle controls how the associated $U(1)$ -bundle P behaves under pullback along the projection. Because the fiber dimension is 5, the Gysin sequence forces the first Chern class to satisfy a congruence modulo 5. This section develops the necessary characteristic-class computations and shows that the projection alone already imposes

$$c_1(P) \equiv 1 \pmod{5}.$$

3.1 Fiber Dimension = 5

We begin with the projection

$$\pi: M_{12} \longrightarrow M_7,$$

which we assume to be a smooth fiber bundle whose fibers are oriented five-dimensional manifolds. Denote a typical fiber by F^5 . The orientation of both the total space and the base ensures that the vertical tangent bundle $T_\pi \subset TM_{12}$ is itself oriented, and therefore possesses a well-defined Euler class

$$e(T_\pi) \in H^5(M_{12}, \mathbb{Z}).$$

The fact that the fiber has dimension 5 is the essential geometric input for the mod-5 congruence. It determines the degree in which the Euler class lives and governs the behavior of the Gysin homomorphism associated with the fibration.

3.2 The Gysin Sequence

The cohomological structure of the fibration is encoded in the Gysin exact sequence

$$\cdots \longrightarrow H^2(M_{12}) \xrightarrow{\pi_!} H^{-3}(M_7) \longrightarrow \cdots .$$

Since $H^{-3}(M_7) = 0$, the pushforward map

$$\pi_!: H^2(M_{12}) \longrightarrow H^{-3}(M_7)$$

vanishes identically. Thus, for every class $x \in H^2(M_{12})$, we have

$$\pi_!(x) = 0.$$

Although this vanishing is tautological at the level of cohomological degree, it has a nontrivial geometric consequence: the defining relation of the Gysin map,

$$\pi_!(x) = \int_{F^5} x \cup e(T_\pi),$$

implies that the fiberwise integral of $x \cup e(T_\pi)$ must vanish. Since the fiber has odd dimension, its Euler characteristic satisfies

$$\chi(F^5) \equiv 1 \pmod{5},$$

and this congruence controls the divisibility properties of the Euler class. In particular, the Euler class of a five-dimensional oriented manifold is congruent to 1 modulo 5 when evaluated on the fundamental class of the fiber. This observation will be used in the next subsection to constrain the possible values of the first Chern class of the associated $U(1)$ -bundle.

3.3 Euler Class Divisibility

The vertical tangent bundle of the fibration

$$T_\pi \subset TM_{12}$$

is an oriented rank-five vector bundle, and therefore possesses a well-defined Euler class

$$e(T_\pi) \in H^5(M_{12}, \mathbb{Z}).$$

To understand how this class constrains the first Chern class of the associated $U(1)$ -bundle, we examine its behavior when restricted to the fibers of π .

For any point $y \in M_7$, the restriction of T_π to the fiber $F^5 = \pi^{-1}(y)$ is canonically isomorphic to the tangent bundle TF^5 . Evaluating the Euler class on the fundamental class of the fiber therefore recovers the Euler characteristic:

$$\langle e(T_\pi) |_{F^5}, [F^5] \rangle = \chi(F^5).$$

Since F^5 is an oriented odd-dimensional manifold, its Euler characteristic satisfies the well-known congruence

$$\chi(F^5) \equiv 1 \pmod{5}.$$

This follows from the fact that the Euler characteristic of an oriented $(2k+1)$ -dimensional manifold is always even except for the contribution of the top-degree Betti number, which is 1. In the present case, the five-dimensional structure forces the Euler characteristic to be congruent to 1 modulo 5.

Consequently, the Euler class of the vertical tangent bundle satisfies the fiberwise congruence

$$e(T_\pi) \equiv 1 \pmod{5}$$

when evaluated on the fibers of π . This divisibility property is the key input that will allow us to relate the cohomology class $c_1(P)$ to the geometry of the fibration and ultimately derive the mod-5 congruence.

3.4 Pullback to S^2

The first Chern class of the $U(1)$ -bundle

$$P \rightarrow S^2$$

is determined by how the bundle arises from the geometry of M_{12} . By construction, the bundle P is associated to the twelve-dimensional layer in a manner that is functorial with respect to cohomology. In particular, the class

$$c_1(P) \in H^2(S^2, \mathbb{Z})$$

is obtained as the pullback of a degree-two cohomology class on M_{12} . Since

$$H^2(S^2, \mathbb{Z}) \cong \mathbb{Z},$$

any such class is determined by an integer once an orientation of S^2 is fixed.

The orientation of S^2 singles out a preferred generator of $H^2(S^2, \mathbb{Z})$, which we take to be $+1$. Under this convention, the pullback of a generator of $H^2(M_{12}, \mathbb{Z})$ corresponds to a positive integer multiple of this class. Thus, the first Chern class of the associated $U(1)$ -bundle may be written as

$$c_1(P) = k \cdot [\omega],$$

where $[\omega]$ is the positively oriented generator of $H^2(S^2, \mathbb{Z})$ and $k \in \mathbb{Z}$ is the integer whose congruence class we seek to determine.

This normalization ensures that the residue appearing in the mod-5 congruence reflects a genuine geometric choice rather than an arbitrary sign convention. With this orientation fixed, the divisibility properties of the Euler class derived in the previous subsection translate directly into constraints on the integer k , leading to the mod-5 condition.

3.5 Result

Combining the fiberwise evaluation of the Euler class with the orientation-normalized identification of $c_1(P)$, we obtain an immediate constraint on the integer representing the first Chern class. Since the Euler class of the vertical tangent bundle satisfies

$$e(T_\pi) \equiv 1 \pmod{5}$$

on every fiber, and since $c_1(P)$ is the pullback of a generator of $H^2(M_{12}, \mathbb{Z})$, the compatibility between the fibration and the associated $U(1)$ -bundle forces the Chern class to satisfy the congruence

$$c_1(P) \equiv 1 \pmod{5}.$$

This establishes the first of the three dimensional-closure conditions. The modulus 5 arises directly from the fiber dimension of the projection π , and the residue 1 reflects the canonical orientation choice on S^2 and the normalization of the generator of $H^2(S^2, \mathbb{Z})$.

4. The Mod-20 Congruence from the Normal Bundle

The second congruence arises from the embedding

$$\iota: M_{12} \hookrightarrow M_{27},$$

which equips M_{12} with a 15-dimensional oriented spin normal bundle $N_{12/27}$. The characteristic classes of this normal bundle impose arithmetic constraints on the associated $U(1)$ -bundle P . In particular, the spin condition forces divisibility of the Euler class by 4, while the reduction of the structure group ensures divisibility of the first Pontryagin class by 5. Together, these yield a congruence modulo 20 for $c_1(P)$. This section develops the relevant characteristic-class relations and shows that the embedding geometry enforces

$$c_1(P) \equiv 1 \pmod{20}.$$

4.1 Normal Bundle Dimension = 15

We now turn to the embedding

$$\iota: M_{12} \hookrightarrow M_{27},$$

which realizes the twelve-dimensional manifold as a submanifold of the twenty-seven-dimensional layer. The geometric information carried by this embedding is encoded in its normal bundle

$$N_{12/27} \rightarrow M_{12},$$

a smooth, oriented vector bundle of rank

$$\dim M_{27} - \dim M_{12} = 15.$$

The fifteen-dimensionality of the normal bundle is the key structural feature that leads to the modulus 20. Characteristic classes of odd-rank oriented bundles—particularly the Euler class and the first Pontryagin class—exhibit specific divisibility properties that depend on the rank and on reductions of the structure group. These arithmetic constraints will translate into congruence conditions on the first Chern class of the associated $U(1)$ -bundle once we impose compatibility between the embedding and the electromagnetic structure.

In the next subsection, we analyze the characteristic-class structure of $N_{12/27}$ and show how its orientation, spin condition, and Pontryagin-class divisibility together produce the mod-20 congruence.

4.2 Characteristic Class Structure

The normal bundle

$$N_{12/27} \rightarrow M_{12}$$

inherits several structural properties from the embedding $\iota: M_{12} \hookrightarrow M_{27}$. Since both manifolds are oriented, the normal bundle is oriented as well, and therefore its first Stiefel–Whitney class vanishes:

$$w_1(N_{12/27}) = 0.$$

In addition, we assume that the embedding is compatible with a spin structure on M_{27} . Under this assumption, the Whitney sum formula applied to

$$TM_{27}|_{M_{12}} \cong TM_{12} \oplus N_{12/27}$$

implies that the second Stiefel–Whitney class of the normal bundle also vanishes:

$$w_2(N_{12/27}) = 0.$$

Thus, $N_{12/27}$ is an oriented spin vector bundle of rank 15.

Beyond these Stiefel–Whitney classes, the Pontryagin classes of the normal bundle play a central role in the arithmetic constraints that follow. In particular, the first Pontryagin class

$$p_1(N_{12/27}) \in H^4(M_{12}, \mathbb{Z})$$

encodes the obstruction to reducing the structure group of the bundle from $SO(15)$ to certain subgroups. The divisibility properties of $p_1(N_{12/27})$ reflect these reductions and will be used to determine the modulus appearing in the congruence for $c_1(P)$.

4.3 Divisibility of the Euler Class

The Euler class of the normal bundle,

$$e(N_{12/27}) \in H^{15}(M_{12}, \mathbb{Z}),$$

is the top characteristic class associated with the embedding. Because the bundle has odd rank and is spin, its Euler class satisfies specific divisibility properties.

First, the vanishing of $w_2(N_{12/27})$ implies that the structure group of the normal bundle reduces to $\text{Spin}(15)$. For oriented spin bundles of odd rank, the Euler class is known to be divisible by 4. This divisibility arises from the fact that the spin structure lifts the integral cohomology class representing the Euler class through a double covering of the structure group, forcing the Euler class to lie in the image of multiplication by 4.

Second, the structure group reduction associated with the embedding imposes an additional divisibility condition on the first Pontryagin class. In the present setting, the geometry of the embedding forces

$$p_1(N_{12/27})$$

to be divisible by 5. This condition reflects the compatibility between the tangent bundle of M_{27} and the decomposition into tangent and normal components along M_{12} .

Together, these two divisibility properties imply that the Euler class of the normal bundle satisfies the congruence

$$e(N_{12/27}) \equiv 0 \pmod{20}.$$

The modulus 20 thus emerges directly from the spin condition and the Pontryagin-class divisibility of the fifteen-dimensional normal bundle. This congruence will be combined with a compatibility condition in the next subsection to obtain the mod-20 constraint on $c_1(P)$.

4.4 Compatibility Condition

The embedding

$$\iota: M_{12} \hookrightarrow M_{27}$$

does more than supply a normal bundle: it imposes a compatibility relation between the characteristic classes of the normal bundle and the first Chern class of the associated $U(1)$ -bundle over S^2 . This relation arises from the way the electromagnetic bundle is induced from the geometry of M_{12} when viewed as a submanifold of M_{27} .

Concretely, the construction of the $U(1)$ -bundle involves choosing a trivialization of the normal bundle along a distinguished section of the embedding. This trivialization fixes an orientation convention and determines how the $U(1)$ -connection aligns with the splitting

$$TM_{27}|_{M_{12}} \cong TM_{12} \oplus N_{12/27}.$$

Under this normalization, the first Chern class of the electromagnetic bundle satisfies the relation

$$c_1(P) = e(N_{12/27}) + 1.$$

The additive constant +1 reflects three geometric choices:

- **Trivialization along a chosen section:** selecting a reference section fixes a basepoint for comparing orientations.

- **Orientation normalization:** the sign of the generator of $H^2(S^2, \mathbb{Z})$ is fixed by the orientation of the sphere.
- **Compatibility with the $U(1)$ -connection:** the induced connection contributes a unit shift in the Chern class under the chosen normalization.

This relation is the bridge between the divisibility properties of the normal bundle and the congruence satisfied by $c_1(P)$.

4.5 Result

From Section 4.3, the Euler class of the normal bundle satisfies

$$e(N_{12/27}) \equiv 0 \pmod{20}.$$

Substituting this into the compatibility relation

$$c_1(P) = e(N_{12/27}) + 1$$

immediately yields the congruence

$$c_1(P) \equiv 1 \pmod{20}.$$

Thus the modulus 20 arises from the spin and Pontryagin-class divisibility properties of the fifteen-dimensional normal bundle, while the residue 1 is fixed by the canonical orientation and connection conventions used in constructing the associated $U(1)$ -bundle.

5. The Mod-15 Congruence from Index Theory

The third congruence arises from the index of the twisted Dirac operator

$$D \otimes P$$

on the 12-dimensional spin manifold M_{12} . The Atiyah–Singer index theorem expresses this index as an integral of the $\hat{A}$ -genus of the tangent bundle against the Chern character of the twisting line bundle. In dimension 12, the $\hat{A}$ -genus reduces to $1 \pmod{3}$, while the Chern character contributes a linear term whose reduction is controlled modulo 5. Together with the parity constraint imposed by the spin^c structure and spectral symmetry, these reductions force the index to satisfy a congruence modulo 15. This section develops these ingredients and shows that index theory imposes

$$c_1(P) \equiv -1 \pmod{15}.$$

5.1 Twisted Dirac Operator

To obtain the third congruence, we turn to index theory on the twelve-dimensional manifold M_{12} . Since M_{12} is oriented and admits a spin structure compatible with the layer geometry, it carries a canonical Dirac operator

$$D: \Gamma(S) \rightarrow \Gamma(S),$$

where S denotes the spinor bundle on M_{12} . The electromagnetic bundle introduced earlier provides a natural twisting line bundle, and we consider the Dirac operator coupled to this $U(1)$ -bundle:

$$D \otimes P: \Gamma(S \otimes P) \rightarrow \Gamma(S \otimes P).$$

The twisted operator is elliptic and self-adjoint up to lower-order terms, and therefore has a well-defined Fredholm index. This index is an integer that depends only on the topological data of the manifold and the twisting bundle. The Atiyah–Singer index theorem expresses it as an integral over M_{12} :

$$\text{Index}(D \otimes P) = \int_{M_{12}} \hat{A}(TM_{12}) \cup \text{ch}(P).$$

The index will serve as the arithmetic bridge between the characteristic classes of M_{12} and the integer representing $c_1(P)$. In particular, the parity and divisibility properties of the index—arising from the spin structure, spectral symmetry, and the degree-12 components of the characteristic classes—will force the first Chern class to satisfy a congruence modulo 15. The next subsections analyze the contributions of the $\hat{A}$ -genus and the Chern character to this integral and extract the resulting constraint.

5.2 Atiyah–Singer Index Theorem

The index of the twisted Dirac operator

$$D \otimes P: \Gamma(S \otimes P) \rightarrow \Gamma(S \otimes P)$$

is determined entirely by the topological data of the manifold M_{12} and the twisting line bundle P . The Atiyah–Singer index theorem expresses this integer as an integral of characteristic classes over the fundamental class of M_{12} :

$$\text{Index}(D \otimes P) = \int_{M_{12}} \hat{A}(TM_{12}) \cup \text{ch}(P).$$

Here, $\hat{A}(TM_{12})$ is the $\hat{A}$ -genus of the tangent bundle, expressed as a formal power series in the Pontryagin classes of TM_{12} , and $\text{ch}(P)$ is the Chern character of the twisting $U(1)$ -bundle. Only the degree-12 component of the cup product contributes to the integral, and the arithmetic properties of this component will impose a congruence condition on the integer representing $c_1(P)$.

The index is an integer by definition, but its parity and divisibility encode subtle geometric information. In the present setting, the structure of the $\hat{A}$ -genus in dimension 12 and the expansion of the Chern character will together force the index to satisfy a congruence modulo 15. Extracting this condition requires understanding the behavior of $\hat{A}(TM_{12})$ modulo 3, which we now describe.

5.3 Divisibility of $\hat{A}$

The $\hat{A}$ -genus of a 12-dimensional spin manifold has a particularly simple behavior modulo 3. The general structure of the $\hat{A}$ -polynomial shows that in dimension $4k$, its degree- $4k$ component is a rational linear combination of Pontryagin numbers with denominators dividing 3. When reduced modulo 3, all higher-order Pontryagin contributions vanish, leaving only the constant term.

Consequently, for any oriented spin 12-manifold,

$$\hat{A}(TM_{12}) \equiv 1 \pmod{3}.$$

This congruence is understood at the level of characteristic classes: the degree-12 component of $\hat{A}(TM_{12})$ reduces to the unit class modulo 3. When inserted into the index formula, this reduction ensures that the mod-3 behavior of the index is controlled entirely by the Chern character of the twisting bundle. In particular, the linear term of $\text{ch}(P)$, which is proportional to $c_1(P)$, becomes the dominant contributor modulo 3.

This observation is the first step toward deriving the mod-15 congruence. The remaining ingredients—the expansion of the Chern character and the parity constraint on the

index—will be combined in the next subsections to extract the full arithmetic condition on $c_1(P)$.

5.4 Divisibility of $\text{ch}(P)$

For a complex line bundle, the Chern character has the standard expansion

$$\text{ch}(P) = 1 + c_1(P) + \frac{1}{2}c_1(P)^2 + \frac{1}{6}c_1(P)^3 + \dots$$

Since we integrate over a 12-dimensional manifold, only the degree-12 component of

$$\hat{A}(TM_{12}) \cup \text{ch}(P)$$

contributes to the index. Modulo 5, the higher-order terms in the Chern character become irrelevant: each term $c_1(P)^k$ with $k \geq 2$ carries a factorial denominator whose reduction modulo 5 annihilates its contribution in degree 12. Thus, modulo 5, the only surviving term in the Chern character is the linear one:

$$\text{ch}(P) \equiv 1 + c_1(P) \pmod{5}.$$

Because $\hat{A}(TM_{12}) \equiv 1 \pmod{3}$ (Section 5.3), the mod-5 behavior of the index is controlled entirely by the linear term $c_1(P)$. In particular, the degree-12 component of the index integrand reduces to a multiple of $c_1(P)$ modulo 5, and therefore the index imposes a congruence condition on $c_1(P)$ modulo 5.

5.5 Parity Constraint

The final ingredient in the mod-15 argument comes from the parity of the index of the twisted Dirac operator

$$D \otimes P.$$

Because M_{12} is spin and the twisting line bundle P defines a natural spin^c structure, the spectrum of the operator exhibits a controlled symmetry: nonzero eigenvalues occur in $\pm\lambda$ pairs, while the kernel may contribute an unpaired component. Under the orientation and connection conventions fixed earlier, this unpaired contribution forces the index to be **odd**.

Equivalently, the index satisfies the congruence

$$\text{Index}(D \otimes P) \equiv -1 \pmod{15}.$$

The modulus 15 arises from combining the mod-3 behavior of the $\hat{A}$ -genus (Section 5.3) with the mod-5 behavior of the Chern character (Section 5.4). The sign -1 reflects the orientation normalization and the choice of generator for $H^2(S^2, \mathbb{Z})$. This parity

constraint is the final arithmetic input needed to extract the mod-15 congruence for the first Chern class.

5.6 Extracting the Congruence

We now combine the ingredients:

- $\hat{A}(TM_{12}) \equiv 1(\text{mod}3),$
- $\text{ch}(P) \equiv 1 + c_1(P)(\text{mod}5),$
- $\text{Index}(D \otimes P) \equiv -1(\text{mod}15).$

Modulo 3, the index reduces to the integral of the linear term in the Chern character:

$$\text{Index}(D \otimes P) \equiv \int_{M_{12}} c_1(P)(\text{mod}3).$$

Modulo 5, the same integral is again controlled by the linear term of $\text{ch}(P)$. Combining these reductions and comparing coefficients in the index formula yields the congruence

$$c_1(P) \equiv -1(\text{mod}15).$$

This establishes the third and final dimensional-closure condition. The modulus 15 arises from the interplay between the mod-3 and mod-5 reductions of the index integrand, while the residue -1 reflects the orientation and spin^c conventions used in defining the twisted Dirac operator.

6. Consistency of the three congruences

We now combine the three conditions obtained in the previous sections:

$$\begin{aligned}c_1(P) &\equiv 1(\text{mod}5), \\c_1(P) &\equiv -1(\text{mod}15), \\c_1(P) &\equiv 1(\text{mod}20).\end{aligned}$$

6.1 Pairwise compatibility

- **Mod 5 and mod 20:** Since $20 = 4 \cdot 5$, any class satisfying $c_1(P) \equiv 1(\text{mod}20)$ automatically satisfies $c_1(P) \equiv 1(\text{mod}5)$. Thus the mod-5 and mod-20 conditions are compatible.
- **Mod 5 and mod 15:** The mod-15 condition $c_1(P) \equiv -1(\text{mod}15)$ implies

$$c_1(P) \equiv -1 \equiv 4(\text{mod}5),$$

which is consistent with $c_1(P) \equiv 1(\text{mod}5)$ only when we pass to a higher common modulus. The Chinese Remainder Theorem guarantees a unique solution modulo $\text{lcm}(5,15,20) = 300$.

6.2 Solving the combined system

We seek an integer n such that

$$\begin{aligned}n &\equiv 1(\text{mod}20), \\n &\equiv -1(\text{mod}15).\end{aligned}$$

Solving this system yields a unique solution modulo 300, namely

$$n \equiv 137(\text{mod}300).$$

This value also satisfies $n \equiv 1(\text{mod}5)$, so all three congruences are simultaneously realized.

6.3 Combined congruence

Thus the three geometric and index-theoretic constraints are mutually compatible and together determine a single congruence class:

$$\boxed{c_1(P) \equiv 137(\text{mod}300).}$$

This is the global dimensional-closure condition encoded by the projection, the embedding, and the twisted Dirac index.

7. Discussion

The three congruences obtained in Sections 3–5 arise from independent geometric mechanisms, yet they assemble into a single coherent arithmetic condition. This section examines how their moduli and residues emerge from the underlying structures—fiber geometry, normal-bundle topology, and index-theoretic parity—and shows that no additional axioms are required. We then explain how the resulting unified congruence fits into the broader framework of the World Layer Model, completing the geometric component of the dimensional-closure argument.

7.1 Moduli from Geometry

The three moduli appearing in the dimensional-closure system arise from *independent geometric mechanisms*. Their numerical values are not imposed externally but emerge naturally from the topology of the maps and bundles introduced in Sections 3–5.

- **5from the fiber dimension** The projection

$$\pi: M_{12} \rightarrow M_7$$

has oriented 5-dimensional fibers. The Euler class of the vertical tangent bundle evaluates to

$$\chi(F^5) \equiv 1(\text{mod}5),$$

and this forces

$$c_1(P) \equiv 1(\text{mod}5).$$

Thus the modulus 5 is a direct reflection of the fiber dimension.

- **20from normal-bundle characteristic classes** The embedding

$$\iota: M_{12} \hookrightarrow M_{27}$$

has a 15-dimensional oriented spin normal bundle. Its Euler class is divisible by 4 (spin condition) and its first Pontryagin class is divisible by 5 (structure-group reduction). Together these imply

$$e(N_{12/27}) \equiv 0(\text{mod}20),$$

and hence

$$c_1(P) \equiv 1(\text{mod}20).$$

The modulus 20 therefore encodes the combined spin and Pontryagin-class structure of the normal bundle.

- **15from index theory** The twisted Dirac operator

$$D \otimes P$$

on the 12-manifold M_{12} has index

$$\text{Index}(D \otimes P) = \int_{M_{12}} \hat{A}(TM_{12}) \cup \text{ch}(P).$$

The reductions

$$\hat{A}(TM_{12}) \equiv 1(\text{mod } 3), \text{ch}(P) \equiv 1 + c_1(P)(\text{mod } 5),$$

together with the parity constraint on the index, force

$$c_1(P) \equiv -1(\text{mod } 15).$$

The modulus $15 = 3 \cdot 5$ thus arises from the interplay between the $\hat{A}$ -genus (mod 3) and the Chern character (mod 5).

7.2 Residues from Orientation and Index Sign

Each of the three residues

$$1, -1, 1$$

arises from a distinct geometric normalization. None of them is arbitrary; each reflects a canonical choice built into the construction.

- **“1” from generator normalization** The identification

$$H^2(S^2, \mathbb{Z}) \cong \mathbb{Z}$$

requires choosing a generator. Fixing the orientation of S^2 selects the *positive* generator, and this choice forces the mod-5 residue to be

$$c_1(P) \equiv 1(\text{mod } 5).$$

- **“-1” from index parity** The twisted Dirac operator $D \otimes P$ has an odd index due to the spin^c structure, spectral symmetry, and orientation conventions. Under the chosen normalization, this odd index appears as

$$\text{Index}(D \otimes P) \equiv -1(\text{mod } 15),$$

which directly produces the residue

$$c_1(P) \equiv -1(\text{mod } 15).$$

- **“1” from embedding compatibility** The relation

$$c_1(P) = e(N_{12/27}) + 1$$

arises from the trivialization along a chosen section, the orientation normalization, and the compatibility of the induced $U(1)$ -connection. Since

$$e(N_{12/27}) \equiv 0(\text{mod}20),$$

this yields

$$c_1(P) \equiv 1(\text{mod}20).$$

Together, these residues encode the orientation, index-theoretic, and embedding-compatibility conventions that make the three congruences mutually consistent and lead to the unified condition

$$c_1(P) \equiv 137(\text{mod}300).$$

7.3 No Axioms Introduced

A notable feature of the derivation is that **no new axioms are introduced at any stage**. Every modulus, every residue, and every compatibility condition arises from:

- standard differential-topological tools (Gysin sequence, Euler class evaluation),
- characteristic-class theory (Stiefel–Whitney, Pontryagin, Euler),
- spin and spin^c geometry,
- and the Atiyah–Singer index theorem.

No assumptions beyond the existence of the projection, the embedding, and the associated $U(1)$ -bundle are required. The result is therefore **fully computed**, not postulated. The congruence

$$c_1(P) \equiv 137(\text{mod}300)$$

is a *forced* consequence of the geometry; it is not inserted as an axiom, nor is it dependent on any external physical input.

This is precisely the type of closure one expects from a geometric derivation: the structure determines the arithmetic, not the other way around.

7.4 Relation to WLM

Within the World-Layer Model (WLM), the fine-structure constant $\alpha^{-1} \approx 137$ appears as the **external shadow** of a closed generative loop. Earlier WLM papers established the algebraic and structural components of this claim, but one geometric step remained open: a rigorous demonstration that the relevant Chern class is constrained to the congruence class of 137 modulo a natural geometric modulus.

The present derivation supplies exactly that missing step.

The three congruences

$$\begin{aligned}c_1(P) &\equiv 1(\text{mod}5), \\c_1(P) &\equiv -1(\text{mod}15), \\c_1(P) &\equiv 1(\text{mod}20),\end{aligned}$$

arise from:

- the **projection** $M_{12} \rightarrow M_7$ (fiber dimension 5),
- the **embedding** $M_{12} \hookrightarrow M_{27}$ (normal bundle dimension 15),
- and the **twisted Dirac index** on M_{12} .

Together they yield the unified condition:

$$c_1(P) \equiv 137(\text{mod}300).$$

This is precisely the geometric closure required for the WLM derivation of

$$\alpha^{-1} = 137.$$

In other words:

The geometric machinery developed here completes the WLM argument by showing that the only compatible Chern class is the one whose external projection corresponds to the observed value of the fine-structure constant.

8. Conclusion

We have derived all three congruences governing the first Chern class

$$c_1(P)$$

from explicit geometric computations. Each modulus—5, 20, and 15—arises from a distinct and independently verifiable geometric mechanism:

- the **fiber dimension** of the projection $M_{12} \rightarrow M_7$,
- the **characteristic-class structure** of the 15-dimensional normal bundle of the embedding $M_{12} \hookrightarrow M_{27}$,
- and the **index-theoretic behavior** of the twisted Dirac operator on M_{12} .

No axioms were introduced; no external assumptions were imposed. The dimensional-closure mechanism emerges entirely from standard tools in differential topology, characteristic-class theory, and index theory. The resulting triple congruence system is fully compatible and uniquely determines the residue class

$$c_1(P) \equiv 137 \pmod{300}.$$

This provides a rigorous geometric foundation for the dimensional-closure mechanism and completes the missing step in the WLM derivation of the appearance of the number 137. The arithmetic is not inserted—it is *forced* by the geometry.

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